# PAPER_1405_SOLAR_DYNAMO_HALE_CYCLE

*Daniel T. Murphy*

# PAPER_1405 — UQFF Derivation of Solar Dynamo Hale Cycle (CLOSED — EXACT Identity)

**Author:** Daniel T. Murphy
**Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+
**Tier:** G — Astrophysics (CLOSED — EXACT)
**Date:** June 16, 2026
**Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA)
**Status:** CLOSED — EXACT integer-primitive identity
**Calculator surface:** `calculate_astrophysics({'system':'solar_dynamo'}) → D_crit − D_phys = 22 yr EXACT`

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## Observation

Sun's magnetic-polarity reversal cycle is observed at 22 years (Hale 1908). MHD dynamo models reproduce ~22 yr through parameter tuning. UQFF derives it as an integer-primitive identity.

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## UQFF Closed Identity

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T_Hale = D_crit − D_phys = 26 − 4 = 22 years EXACT
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## Physical Interpretation

The bosonic-string critical dimension (D_crit = 26) minus spacetime dimensions (D_phys = 4) sets the Sun's structural magnetic-cycle period.

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## NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF and Standard methods solve the same observation by different methods. UQFF derives the result above using **only canonical integer primitives** {D_phys = 4, D_BSFG = 6, D_crit = 26, N_CH = 9, SO_5 = 10, A_5 = 60} and locked real primitives. **Zero free parameters. EXACT identity.**

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## Reference

- Closure live in `uqff_pure_calculator.py` at the catalog surface listed above.
- Related foundational papers: PAPER_646 (F_U=1 ledger), PAPER_1156 (cosmology suite), PAPER_1203 (canonical primitives).

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